

Sparsity on $\rho_i = \ell_i^{-2}$

$\rho_i \rightarrow 0$: the component flattens



$\ell_i \rightarrow \infty$: the same draw, stretched

Sparsity on a_i^2

$a_i^2 \rightarrow 0$: the component vanishes



$a_i \rightarrow 0$: the same draw, scaled down

Additive

$$k(x, x') = \sum_i k_i(x_i, x'_i)$$

first-order effects only



additive • lengthscale

$$\sigma_f^2 \sum_i \tilde{k} \ell_i$$

unnormalised; $\rho_i \sim \text{HC}(\tau)$

additive • amplitude

$$\sum_i a_i^2 \bar{k}_i$$

$\bar{k}_i = \tilde{k}_{\ell_i} / \nu(\ell_i)$; $a_i^2 \sim \text{HC}(\tau)$

the proposal

Product

$$k(x, x') = \prod_i k_i(x_i, x'_i)$$

interactions representable



product • lengthscale

Matérn-5/2 ARD

$\rho_i \sim \text{HC}(\tau)$

SAASBO

product • amplitude

$$\prod_i (1 + a_i^2 \bar{k}_i)$$

$a_i^2 \sim \text{HC}(\tau)$